

When and why randomized exploration works

Author

Presenter: Max Van Fleet

University of North Carolina at Chapel Hill School of Data Science and Society

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Why do we care about Bandits?

- Bandits are the simplest forms of RL and the ideas for solving bandits reappear in Markov Decision processes and reinforcement learning algorithms
- Allow a for a simple environment to explore issues in RL like exploration vs exploitation trade off

What is a Linear Bandit

- Let there be a fixed unknown reward vector $\theta^* \in \mathbb{R}^d$
- Let the action $\mathcal{A}$ be a subset of the $\mathbb{B}_2^d$ i.e for all arms $a \in \mathcal{A}$ we have $\|a\|_2 \leq 1$
- For each time step $1, \dots, T$ pick $a_t \in \mathcal{A}$ and observe reward $y_t = \langle a_t, \theta_* \rangle + \epsilon_t$ where ϵ_t is zero mean subgaussian noise
- Let $a_* = \arg \max_{a \in \mathcal{A}} \langle a, \theta_* \rangle$
- The goal is to minimize cumulative pseudo regret defined by $\sum_{i=1}^T \langle a_*, \theta_* \rangle - \langle a_i, \theta_* \rangle$

2 Approaches

- Optimism Approaches
 - Use the principle of Optimism in the Face of Uncertainty to balance exploration and exploitation
 - not great for implementation because it involves solving a bi-level optimization problem $(a_t, \theta_t) \in \arg \max_{(a, \theta) \in (\mathcal{A}, \Theta_{t-1})} \langle a, \theta \rangle$
- Randomized Exploration(Thompson Sampling)
 - rather than optimizing over all feasible reward vectors randomly choose a reward vector from a confidence set and optimize with respect to that
 - Random draws from the posterior much faster
 - Previous analysis fall back on the idea of optimism to produce bounds on regret [Agrawal and Goyal, 2012]
 - Better empirical performance as seen in [Chapelle and Li, 2011]

Empirical Performance of Thompson Sampling

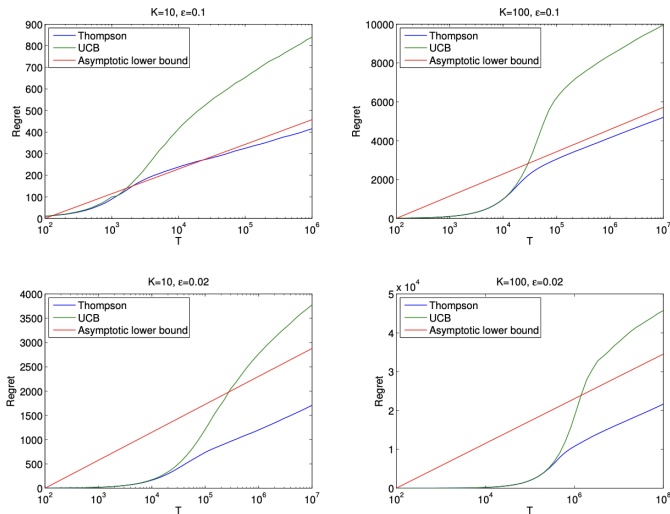


Figure 1: Cumulative regret for $K \in \{10, 100\}$ and $\epsilon \in \{0.02, 0.1\}$. The plots are averaged over 100 repetitions. The red line is the lower bound (2) shifted such that it goes through the origin.

Linear Thompson Sampling Lin TS

Algorithm 1 Linear Thompson sampling (LinTS)

Require: Inflation parameter ι .

- 1: Initialize $\Sigma_1 \leftarrow \lambda \mathbb{I}$ and $\hat{\Theta}_1 \leftarrow 0$
 - 2: **for** $t = 1, 2, \dots$ **do**
 - 3: Observe $\mathcal{A}_t$
 - 4: Sample $\tilde{\Theta}_t \sim \mathcal{N}(\hat{\Theta}_t, \iota^2 \Sigma_t)$
 - 5: $\tilde{A}_t \leftarrow \arg \max_{A \in \mathcal{A}_t} \langle A, \tilde{\Theta}_t \rangle$
 - 6: Observe reward Y_t
 - 7: $\Sigma_{t+1}^{-1} \leftarrow \Sigma_t^{-1} + \tilde{A}_t \tilde{A}_t^\top$
 - 8: $\hat{\Theta}_{t+1} \leftarrow \Sigma_{t+1}^{-1} (\Sigma_t^{-1} \hat{\Theta}_t + \tilde{A}_t Y_t)$
 - 9: **end for**
-

[Hamidi and Bayati, 2023]

- V_t is the posterior covariance of θ
- 4. There is a confidence ellipsoid centered around the best estimate of θ_* inflated in order to explore sufficiently
- 7. Do a rank 1 update of the posterior covariance of θ
- 8. We update our best guess of θ_* through a ridge regularization

Two types of regret bounds

- Frequentist Regret

- $\mathbb{E}[\sum_{i=1}^T \langle a_*, \theta_* \rangle - \langle a_i, \theta_* \rangle]$

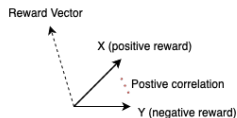
- Bayesian Regret

- $\mathbb{E}_{\theta \sim Q}[\sum_{i=1}^T \langle a_*, \theta_* \rangle - \langle a_i, \theta_* \rangle]$

- the bayesian regret averages the frequentist regret over the prior Q on θ

- Bayesian regret is less informative than frequentist regret, a bound on the Bayesian regret does not imply meaningful bound on the pseudo regret [Lattimore and Szepesvari, 2017]

[Hamidi and Bayati, 2023] Example 2 Intuition



- Consider an action set with $\mathcal{A} = \{\mathbf{0}, X, Y\}$, $\|\theta_*\| \approx \sqrt{d}$
- Arm X has positive reward, Y has negative reward
- Arm X is positively correlated with arm Y i.e the angle between the two is acute
- We will start with a prior mean of $\mathbf{0}$, however the true parameter has a mean shift of $\theta_* \sim \mathcal{N}(\mu\mathbf{1}, \sigma^2 I)$, $\mu > 0$
- If $a_1 = Y$ the negative reward makes $\langle \hat{\theta}_2, X \rangle, \langle \hat{\theta}_2, Y \rangle < 0$ so we $a_2 = \mathbf{0}$ with high probability
- $V_{t+1}^{-1} = V_t^{-1} + \mathbf{0}\mathbf{0}^T \Rightarrow$ no update to the covariance matrix so we go back to $\mathbf{0}$ with high probability
- Thompson sampling basically gets stuck and with high probability will choose the arm at the origin, incurring linear regret

Exact Proposition from [Hamidi and Bayati, 2023]

Proposition 3.2 For fixed $\mu, \sigma > 0$, and for sufficiently large d , we have

1. $\langle \Theta^*, A' \rangle \leq -\frac{1}{2}\mu\sqrt{d} \leq \frac{1}{2}\mu\sqrt{d} \leq \langle \Theta^*, A \rangle$, with probability at least $\frac{7}{8}$.
2. $\tilde{A}_1 = A'$ with probability $\frac{1}{4}$.
3. Conditional on $\tilde{A}_1 = A'$, $\langle \hat{\Theta}_2, A \rangle$ and $\langle \hat{\Theta}_2, A' \rangle$ are less than $-\frac{1}{8}\mu\sqrt{d}$, with probability at least $\frac{15}{16}$.
4. Conditional on $\tilde{A}_1 = A'$, $\tilde{A}_2 \neq 0$ with probability at most $\exp(-\Omega(d))$.
5. For all $T \leq \exp(\Omega(d))$, $\text{Regret}(T, \pi^{\text{LinTS}}) \geq \Omega(T\sqrt{d})$.

Notation:

- $\mathcal{A} = \{0, A', A\}$

How do they fix it

- We can inflate the variance of the confidence ellipsoid which forces us to explore more but this yields an extra $\sqrt{d}$ term to our regret
- [Hamidi and Bayati, 2023] propose that rather than having the inflated variance over the whole course of the algorithm if the posterior covariance ellipsoid becomes too thin then inflate the variance
- [Abeille et al., 2025] We can make additional assumptions on the structure of the problem and keep the variance uninflated, while achieving mini-max optimal regret bounds

- Pick $a_t, \theta_t : \langle a_t, \theta_t \rangle \geq \langle a', \theta' \rangle, \forall (a', \theta') \in \mathcal{A} \times \Theta_{t-1}$, solve via bilinear optimization
- Regret $R_T \leq d\sqrt{T} \log T$
- proof:

Decompose single round regret:

$$r_t = \langle a_*, \theta_* \rangle - \langle a_t, \theta_t \rangle + \langle a_t, \theta_t - \theta_* \rangle$$

First two terms fall out do to optimism

$$\langle a_*, \theta_* \rangle - \langle a_t, \theta_t \rangle \leq 0$$

Third term we

$$\begin{aligned} r_t &= \langle a_t, \theta_t - \theta_* \rangle \\ &\leq \|a_t\|_{V_t} \\ &\leq 2\beta \|a_t\|_{V_t} \end{aligned}$$

finish with elliptical potential lemma

Variance Inflated Lin TS

- Sample $\tilde{\theta}_t \sim \Theta_{t-1}^\gamma$
- play $a_t = a^*(\tilde{\theta}_t)$
- $\gamma \approx d \Rightarrow R_t \leq d^{\frac{3}{2}} \sqrt{T} \log(T)$
- proof:

Define the support of the arm set as $J(\theta) = \max_{a \in \mathcal{A}} \langle a, \theta \rangle$

Decompose single round regret $r_t = J(\theta_*) - J(\tilde{\theta}) + \langle a_t, \theta_* - \tilde{\theta}_t \rangle$

Third term treated exactly as Lin UCB $\langle a_t, \theta_* - \tilde{\theta}_t \rangle \leq \sqrt{\gamma} \beta_t \|a_t\|_{V_t^{-1}}$

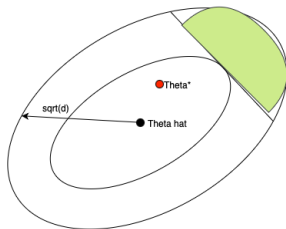
Let $\theta_- = \arg \min_{\theta \in \Theta_{t-1}} J(\theta)$

Let $p \leq \mathbb{P}_{t-1}\{J(\tilde{\theta}_t) \geq J(\theta_*)\}$ the probability of optimism

Now bounding the first two terms

$$\begin{aligned} J(\theta_*) - J(\tilde{\theta}_t) &\leq J(\theta_*) - J(\theta_-) \\ &\leq \frac{1}{p} \mathbb{E}[J(\theta_t) - J(\theta_-); J(\tilde{\theta}_t) \geq J(\theta)] \\ &\leq \frac{\sqrt{\gamma}}{p} \beta_t \|a_t\|_{V_t^{-1}} \end{aligned}$$

Variance Inflated Thompson Sampling continued



Origin ●

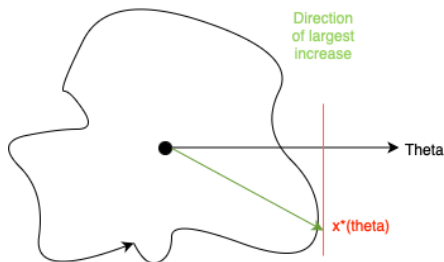
- We enforce optimism by inflating the confidence ellipsoid
- $\gamma = d \Rightarrow$ constant volume fraction $\Rightarrow p \geq \text{constant}$
- with high probability we know that θ_* is in the inner ellipsoid so if we sample from a variance inflated ellipsoid the green area will represent the optimism
- Does not tell us anything about why randomized exploration works only says that if we sample from a sufficiently large confidence ellipsoid we will have some constant probability of optimism and

When and why randomized exploration works

[Abeille et al., 2025]

- Smoothness Condition
 - Exploitation is easy if $\text{regret}(\theta + \delta) \approx \text{regret}(\theta)$
 - If your guess of θ_* is anywhere near the θ_* you will approximately get optimal regret
- Strong Convexity Condition
 - Exploration is easy if $a_*(\theta + \delta) \not\approx a_*(\theta)$
 - if you perturb θ by a little then the optimal arms change a lot
- Main idea is that to balance exploration and exploitation you can explore a lot because small changes in θ cause big changes in the arm you choose while keeping regret very small

Convexity condition



- Recall $J(\theta) = \max_{a \in \mathcal{A}} \langle a, \theta \rangle$
- $a^*(\theta) = \nabla J(\theta)$ almost everywhere
- Slide a line perpendicular to θ and find the point on boundary of $\mathcal{A}$
- in order to make J increase the most you would move along the a vector therefore $\nabla J(\theta) = a^*(\theta)$
- If we want to make sure that the optimal action is changed if we perturb θ we might consider $\|\nabla J(\theta) - \nabla J(\theta')\| \geq m \|\theta - \theta'\|$

Smoothness Condition

$$\begin{aligned} r_t &= J(\theta_*) - \langle a_t, \theta_* \rangle \\ &= J(\theta_*) - J(\tilde{\theta}_t) + J(\tilde{\theta}_t) - \langle a_t, \theta_* \rangle \\ &= J(\theta_*) - J(\tilde{\theta}_t) - \langle \nabla J(\tilde{\theta}_t), \theta_* - \tilde{\theta}_t \rangle \\ &= D_J(\theta_*, \tilde{\theta}_t) \end{aligned}$$

Where $D_J(\theta_*, \tilde{\theta}_t)$ is the bregman divergence

For our smoothness condition we ask that $D_J(\theta_*, \tilde{\theta}_t) \leq M \|\theta_* - \tilde{\theta}_t\|^2$

Finalizing Smoothness and Convexity condition

- $\|\nabla J(\theta) - \nabla J(\theta')\| \geq m\|\theta - \theta'\|$
- The first term is scale invariant and the second term is homogenous therefore there is no fixed m that works because we can change the scale of θ
- To fix this $\nabla J^2(\theta) = 2J(\theta)\nabla J(\theta)$, we get that $J(\theta)$ is homogenous
- Altogether $\|2J(\theta)\nabla J(\theta) - 2J(\theta')\nabla J(\theta')\| \geq m\|\theta - \theta'\|$

Assumptions

- we consider the class of algorithms where $\theta_t = \hat{\theta}_{t-1} + V_{t-1}^{-1/2} \eta_t$
- **Assumption 1:** The perturbations $(\eta_t)_{t \geq 1}$ are independent rotationally invariant random variables for which there exists a constant $K > 0$ such that

$$1 \leq \mathbb{E} \langle u, \eta_t \rangle^2 \leq K^2 \text{ and } \mathbb{E} \langle u, \eta_t \rangle^4 \leq K^4, \forall u \in \mathbb{S}^{d-1}, t \geq 1$$

- rotationally invariant implies we get unique actions as draws are diffuse, and bounded moments ensure that the explorations is isotropic and controlled
- **Assumption 2:** The action set $\mathcal{A}$ is a closed absorbing subset of $\mathbb{B}_2^d$ and there exists a norm $\|\cdot\|_*$ and constants $M, m > 0$ such that J^2 is m -strongly convex and M -Smooth

$$\frac{m}{2} \|\theta - \theta'\|_*^2 \leq D_{J^2}(\theta, \theta') \leq \frac{M}{2} \|\theta - \theta'\|_*^2$$

Setup of proof of main result

- notation: $f(x) \gtrsim g(x)$ if $f(x) = O(g(x))$
- We will work on the $1 - \delta$ event where $\theta_* \in \cap_{t \geq 1} \Theta_{t-1}$
- Confidence set widths: $\beta = R\sqrt{\lambda} + S\sqrt{2\log(1/\delta) + \log(\det(V_t)/\lambda^d)}$
- We define $\bar{R}_T := \sum_{t=1}^T \mathbb{E}_{t-1} r_t$
- Using a martingale difference we can find $R_T \lesssim \bar{R}_T + \sqrt{dT \log(\frac{dT}{\delta})}$
- Our first step will be to decompose the regret

Regret Decomposition

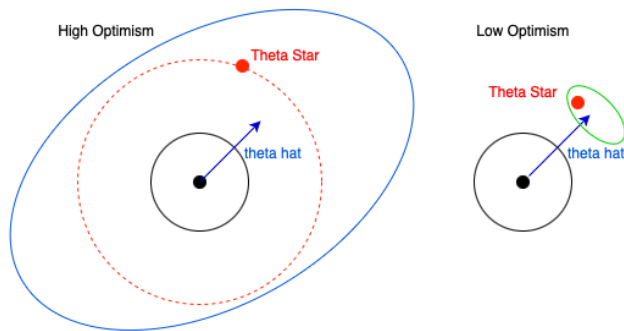
- let p_{t-1} be the conditional probability of optimism i.e $\mathbb{P}_{t-1}\{J(\theta_t) \leq J(\theta_*)\}$, let $\chi_{t-1} = \mathbf{1}[p_{t-1} \leq p]$, they set $p = \frac{1}{16K^4}$ where K is constant from assumption 1

$$\begin{aligned}\bar{R}_T &= \sum_{t=1}^T \chi_{t-1} \mathbb{E}_{t-1} r_t + \sum_{t=1}^T (1 - \chi_{t-1}) \mathbb{E}_{t-1} r_t \\ &\lesssim \frac{M(\beta \vee K^2 d)}{J(\theta_*)} \sum_{t=1}^T \chi_{t-1} \sup_{u \in \mathbb{B}_2^d} \|V_{t-1}^{-1/2} u\|_*^2 + \\ &\quad K^4 (\beta_n \vee K\sqrt{d}) \sum_{t=1}^T \mathbb{E}_{t-1} \|a_t\|_{\Theta_{1:t-1}}\end{aligned}$$

The second term is optimistic and can be bound via a concentration lemma and the elliptical potential lemma

$$\text{second term} \lesssim (\beta_n \vee K\sqrt{d}K^4 \sqrt{T(d \log(1 + T/(d\lambda)) + \log(1 + \delta))})$$

High and Low optimism



Basically the idea is to decompose the regret into two scenarios the first is the optimistic time steps where our confidence ellipsoid is large leaving us with lots of "free optimism" and then a no optimism phase where $\|\hat{\theta}\| \approx \|\theta_*\|$

High probability optimism regret decomposition

p_{t-1} is conditional probability of optimism, for almost every $\theta_t, \theta'_t \in \mathbb{R}^d$ such that $J(\theta') \geq J(\theta)$

Use Convexity, cuachy-schwarz and optimism to rewrite $D_J(\theta_*, \theta_t)$

$$D_J(\theta_*, \theta_t) \leq \|\nabla J(\theta'_t)\|_{V_{t-1}^{-1}} \|\theta'_t - \theta_t\|_{V_{t-1}} + \|\nabla J(\theta_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta_t\|_{V_{t-1}}$$

$$\text{Let } Q(X) = \begin{cases} \frac{1}{p_{t-1}} P_{t-1}(X \cap \{\theta \in \mathbb{R}^d : J(\theta) \geq J(\theta_*)\}), & p_{t-1} \neq 0 \\ \text{any arbitrary measure} & \text{otherwise} \end{cases}$$

Since for almost all $\theta'_t \in \mathbb{R}^d$, $J(\theta') \geq J(\theta)$ and Q is a diffuse measure on X we get

$$\begin{aligned} \mathbb{E}_{t-1} D_J(\theta_*, \theta_t) &\leq \int \|\nabla J(\theta'_t)\|_{V_{t-1}^{-1}} \|\theta'_t - \theta_t\|_{V_{t-1}} (P_{t-1} \otimes Q)(d\theta_t \times d\theta'_t) \\ &\quad + \int \|\nabla J(\theta_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta_t\|_{V_{t-1}} P_{t-1}(d\theta_t) \end{aligned}$$

Solving the integrals and using $\nabla J(\theta) = a_t$ to get

$$\mathbb{E}_{t-1} D_J(\theta_*, \theta_t) \leq \frac{4\tilde{\beta}_{t-1}}{p_{t-1}} \mathbb{E}_{t-1} [\|a_t\|_{V_{t-1}^{-1}}^2]^{\frac{1}{2}}$$

low probability optimism regret decomposition

Use law of cosines convexity and cauchy-schwarz to rewrite

$$D_J(\theta_*, \theta'_t) \leq \frac{1}{J(\theta_*)} D_{J^2}(\theta_*, \theta'_t) + \|\nabla J(\theta'_t)\|_{V_{t-1}} \|\theta_* - \theta'_t\|_{V_{t-1}}$$
$$D_J(\theta_*, \theta_t) \leq D_J(\theta_*, \theta'_t) + \|\nabla J(\theta'_t) - \nabla J(\theta_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta_t\|_{V_{t-1}}$$

Combining the above two bounds for all $\theta_t, \theta'_t \in \mathbb{R}^d, J(\theta'_t) \leq J(\theta_*)$

$$D_J(\theta_*, \theta_t) \leq \frac{1}{J(\theta_*)} D_{J^2}(\theta_*, \theta'_t) + \|\nabla J(\theta_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta_t\|_{V_{t-1}} +$$
$$\|\nabla J(\theta'_t)\|_{V_{t-1}^{-1}} [\|\theta_* - \theta_t\|_{V_{t-1}} + \|\theta_* - \theta'_t\|_{V_{t-1}}]$$

We using the same definition from above for Q we get

$$\begin{aligned} \mathbb{E}_{t-1} D_J(\theta_*, \theta_t) &\leq \frac{1}{J(\theta_*)} \int D_{J^2}(\theta_*, \theta'_t) Q d(\theta'_t) \\ &+ \int \|\nabla J(\theta'_t)\|_{V_{t-1}^{-1}} Q(d\theta'_t) \int \|\theta_* - \theta_t\|_{V_{t-1}} P_{t-1}(d\theta_t) \\ &+ \int \|\nabla J(\theta'_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta'_t\|_{V_{t-1}} Q(d\theta'_t) \\ &+ \int \|\nabla J(\theta_t)\|_{V_{t-1}^{-1}} \|\theta_* - \theta_t\|_{V_{t-1}} P_{t-1}(d\theta_t) \end{aligned}$$

Confidence widths to amount of exploration

- We want to show that V_t grows sufficiently fast with respect to the geometry induced by $\|\cdot\|_*$
- we do this with the change of geometry lemma by relating $\|V_{t-1}^{-1/2}u\|_*$ to the amount of exploration in the direction $u \in \mathbb{B}_2^d$

$$\frac{1}{J(\theta_*)} \|V_{t-1}^{-1/2}u\|_*^2 \lesssim \frac{K}{m} \|\mathbb{E}_{t-1}[a_t a_t^T]^{1/2} V_{t-1}^{-1/2}\|$$

- If $\mathcal{A} = \mathbb{B}_2^d$ we have $m = 1$ for $\|\cdot\|_* = \|\cdot\|_2$ and no change of geometry is needed so we have

$$\begin{aligned} \mathbb{E}_{t-1} a_t a_t^T &= \mathbb{E}_{t-1} [\theta_t \theta_t^T / \|\theta_t\|^2] \approx \frac{1}{\|\theta_*\|^2} \mathbb{E}_{t-1} \theta_t \theta_t^T \\ &\geq \frac{1}{\|\theta_*\|^2} \text{Var}_{t-1} \theta_t = \frac{K^2}{J^2(\theta_*)} V_{t-1}^{-1} \end{aligned}$$

- Final part of the proof is to show relation between the sum of conditional expected increments and their realisation
- For any $r \in (0, 1]$ and $\delta \in (0, 1)$ with probability at least $1 - \delta$ for all $T \geq 1$ and all $u \in r\mathbb{B}_2^d, \omega = d \log(20d^3 n^2 r / \delta^2)$

$$u^T \sum_{t=1}^T a_t a_t^T u + J^2(u) \omega + 5 \geq \frac{1}{2} \sum_{t=1}^T u \mathbb{E}_{t-1}[a_t a_t^T] u$$

- A standard chernoff inequality could be used however in the setting where $J(u) \neq \|u\|$ it is important to obtain $J^2(u)$ dependence which is what the lemma above provides

Putting it all together

- We use the change of geometry lemma and the lemma from the previous slide to bound the regret of the non optimistic time steps
- We can show $\forall t \geq 2, u \in \mathbb{B}_d^2$,
$$\frac{1}{J(\theta_*)} \|V_{t-1}^{-1/2} u\|_*^2 \lesssim \frac{M\omega_{t-1}K^2J(\theta_*)}{m^2N_{t-1}} + \frac{K}{m\sqrt{N_{t-1}}}$$
- Where N_n is the number of steps up to n with low optimism
- this claim allows us to bound the low optimism regret rounds

$$\begin{aligned}\bar{R}_n^{TS} &\lesssim \sqrt{d} + (\beta_T^2 \vee K^2 d) \int_1^n \left[\frac{M^2 \omega_t K^2 J(\theta_*)}{m^2 N_t} + \frac{K}{m\sqrt{N_t}} \right] dt \\ &\lesssim \sqrt{d} + d\sqrt{d}(\beta_T^2 \vee K^2 d) K^2 \frac{M^2}{m^2} \log(dT/(\delta\lambda)) \log(T) + \frac{M}{m} K(\beta_T^2 \vee K^2 d)\end{aligned}$$

First two terms our lower order and that completes the bound

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